University of Florida
College of Public Health & Health Professions Syllabus
Spring 2026 PHC 6053: Regression Methods for the Health and Life Sciences (3 credits)
Delivery Format: Online in E-Learning using CANVAS: http://elearning.ufl.edu/
Note: It is very important to review the course website several times each week and read all announcements carefully.

STARTING THE COURSE: Read this syllabus. Review the E-Learning home page and weekly schedule. Additional guidance for getting started is provided in the E-learning site in CANVAS.

INSTRUCTOR:	Dr. Lixia Wang
Office:	CTRB 5212
Phone Number:	352-294-5919
Email Address:	lixia.wang@ufl.edu
Office Hours/Synchronous Class Meeting:	Mondays 4:00-5:30 PM (ET); Wednesdays 12:00-1:30 PM (ET)
Zoom link to office hours:	check in Modules on E-learning course website

PREFERRED COURSE COMMUNICATIONS:

- The best way to discuss course material or SAS code related questions is attending office hours on Zoom.
- For questions or personal matters that is not appropriate ask in Zoom, please email through CANVAS inbox in E-Learning.
- For more general questions that other students may also have, please use the discussion boards.

TEACHING ASSISTANT: TBD Email via E-Learning in Canvas INBOX

PREREQUISITE: PHC 6052: Introduction to Biostatistical Methods. Students must have prior experience with basic data entry and analysis in SAS. Students who have not taken the pre-requisite course must illustrate their SAS skills at the PHC 6052 level in order to obtain instructor approval to enroll. All students must have access to SAS 9.4.

See https://software.ufl.edu/student-agreements/ and click on "SAS Licensing Agreement" for SAS program purchase information and online documents. To purchase SAS you must be able to come to campus to obtain the software. Alternatively you can use SAS 9.4 on the UFApps system. See http://info.apps.ufl.edu/. You can also use SAS OnDemand for Academics, another free option provided by SAS. See https://www.sas.com/en_us/software/on-demand-for-academics.html.

Although most statistical analyses will be conducted using software SAS in this course, **students should be comfortable working with equations and performing basic mathematical calculations including order of operations, fractions, square roots, logarithms (base *e*), and exponentials (e^x)**.

Note: in statistics the notation **log** is equivalent to the **natural logarithm ln**. This can be confusing as in algebra a log with no base assumed a base of 10. **In this course, a log with no base is assumed to be the natural logarithm, log = ln = $\log_e$.**

IMPORTANT: The textbook uses STATA, but **in PHC 6053 we only use SAS.**

PURPOSE AND OUTCOME

Course Overview: This course introduces graduate students in fields other than statistics to a wide range of modern regression methods. Emphasis is on modeling driven by actual data from studies in a variety of areas, primarily from health, biology, and ecology. The primary topics are multiple linear regression, logistic regression, and Poisson regression. A main goal is to learn what approach to use among the linear and nonlinear models, and how to determine if the fit is adequate. By the end of the course, students will achieve competency in carrying out the analyses in SAS.

Course Objectives: Upon completion of the course, students will be able to:

- CO-1: Select appropriate methods for a scenario; determine if a linear or a nonlinear approach is appropriate
- CO-2: Use statistical software for performing regression analysis in the SAS language
- CO-3: Test and interpret linear models for continuous outcome data (normal linear model)
- CO-4: Test and interpret models for categorical outcome data (logistic and Poisson regression)
- CO-5: Draw appropriate conclusions for both randomized designed experiments and observational studies

- CO-6: Communicate clearly to subject matter experts the purposes and results of complex statistical analysis, both orally and in writing.

Relation to Program Outcomes: This three-credit course is a required concentration core course for MPH Biostatistics students and covers the following MPH Biostatistics competencies.

- Describe the role of biostatistics in public health research.
- Interpret and critique analyses found in public health studies.
- Use appropriate statistical methodology to address public health problems.
- Develop presentations based on statistical methods and analyses for both public health professionals and educated lay audiences.
- Apply software to conduct statistical analyses.

Instructional Methods: This course is presented as an online course. Students will review online materials each week as directed on the E-Learning website for this course. Students can receive feedback and assistance via the course discussion board or email in E-Learning.

My office hours will be held on Zoom. Monday sessions will be used to answer questions regarding the course materials, while Wednesday sessions will focus on going through examples using SAS. You are highly encouraged to attend, although attendance is optional.

Students are expected to engage with the weekly online content listed on the E-Learning wet site. Feedback and assistance are available through the discussion board and Canvas Inbox messages. Taking ownership of your learning is essential. Please proactively seek help form me, our TA, and your peers whenever you encounter difficulties. I am always available to offer guidance on how to make the most effective use of the course resources.

DESCRIPTION OF COURSE CONTENT, COURSE MATERIALS, AND TECHNOLOGY

Required Textbook: *Regression methods in Biostatistics*: Linear, Logistic, Survival, and Repeated Measures Models, 2nd edition (2012), by Vittinghoff, Glidden, Shiboski, and McCulloch. Springer. Book website: http: //www.biostat.ucsf.edu/vgsm. The textbook is available online through the UF library for free as a pdf. A PDF file will also be provided through the E-Learning site for this course. This textbook uses STATA. You can ignore any instruction in the textbook that is directly related to STATA.

Optional Textbook: *Applied Medical Statistics Using SAS* (2013), by Geoff Der and Brian S. Everitt. CRC Press, Boca Raton, FL. ISBN 978-1439867976. The digital version of this book is also available for free from UF library.

Statistical Software – SAS

Analysis will be performed with the SAS software. A SAS information sheet and a list of SAS skills learned in PHC 6052 will be provided on the Canvas course website. Sample SAS code will be provided for all skills required for this course.

Some students find SAS to be the most fun part of the course, others dislike learning the software. Regardless, the MPH program (which operates this course) requires SAS to be an integral part of our instruction. If you have questions about SAS, ask on the discussion board. Do not allow yourself to waste time working in the software, if you are having issues, let us know immediately and we will help as soon as possible. Try to make sure as much of your time as possible in the software is productive. If you find SAS to be fun and are willing to provide moral or tutorial support for other students, feel free to post on the discussion board or let me know!

If you would like to have more reference to further enhance your SAS programming skills, you may consider the books below among many other options:

- *The Little SAS Book: A Primer*, 6th ed. (2019), by Lora Delwiche and Susan Slaughter. SAS, Cary, NC. ISBN 978-1642952834.
- *Applied Statistics and the SAS Programming Language* (2005), by Ron P. Cody and Jeffrey K. Smith.

Assigned Course Materials should be completed during the week assigned. Any questions you have regarding material presented should be clarified by posting on the course discussion board. YOU ARE RESPONSIBLE FOR ALL MATERIAL IN THESE ASSIGNMENTS. Take careful notes including both the specific details and the general concepts discussed.

Videos: If the text in the video is too blurry, try increasing the quality of the YouTube video using the small gear icon which appears at the bottom of the video when it is playing. If you want to view the video faster or slower, you can adjust the speed using the gear icon. Videos in this course may or may not have transcripts or captions.

E-Learning: All course materials are available online through the E-Learning site for this course together with grades, assignments, discussion boards, and other course information. It is very important to check the home page of this site every day and review all announcements carefully. E-learning is accessible at http://elearning.ufl.edu/ or through my.ufl.edu. You must have a valid Gatorlink ID and password. For assistance, call the UF Help Desk at 392-HELP.

Check This Link for Non-SAS Technical Help: Information on many common issues can be found in our e-Learning Support pages at http://studentlife.online.mph.ufl.edu/

For **technical support for our E-learning in Canvas site**, activities and assessments, please contact the Online Course Coordinator at:

Truly Hardemon, MEd
Room Number: HPNP 4173
Phone Number: 352-273-5822
Email Address: **hardemont@ufl.edu**
Office Hours: 8:00 am to 5:00 pm Monday - Friday

For **technical difficulties with E-learning in general** please contact the **UF Help Desk** at:

Learning-support@ufl.edu
(352) 392-HELP - select option 2

Browser Recommendations: https://kb.helpdesk.ufl.edu/FAQs/SupportedBrowsersForUFWebsites
Response Time: For questions asked on Monday - Thursday we will do my best to address all questions completely within 24 hours. For questions asked on Friday, Saturday, or Sunday, we may not be able to respond until Monday the following week.

Announcements: Class Announcements will be sent via the Announcements tool in E-learning, depending on your CANVAS notification settings, you may or may not be notified through your UF email. **Be sure to read all announcements carefully**. **You are responsible for all information in these announcements.** As a student at the University of Florida, it is very important to check your UFL email address and course sites regularly. An easy way to access your UF email account is at https://webmail.ufl.edu/.

Discussion Boards: Reviewing the discussion posts of other students and posting your own can be very helpful.

ACADEMIC REQUIREMENTS AND GRADING

Assignments: Individual assignments will involve either data analysis using SAS or interpretations of SAS outputs. These assignments generally require extended work and should be started as early as possible to allow for any questions or issues that may arise. Software assignments: One opportunity for remediation is allowed during the open window, with a 10%-point deduction. Final grade for each software assignment will be the average of all attempts. Interpretation assignments: **Cannot** be remediated.

Discussion: Discussion will be monitored, and your grade will reflect both adherence to the instructions and the quality of your posts. For each module/unit, you are required to

- Make one original post following the instructions provided in the discussion assignment.
- Respond to at least one post from another participant, also according to the instructions.

Final Interpretation Project: There will be a final project consisting of student illustration of numerous skills covered in this course. The goal of the project is to conduct regression analyses using linear and logistic regression including exploratory data analysis, unadjusted estimates from simple regression models, and a final model.

All coursework must be submitted via E-Learning by the due date and time.

Grading

Requirement	% of final grade
Software Assignments (5)	25%
Interpretation Assignments (5)	40%
Engagement (6)	15%
Final Interpretation Project	20%

The numerical final score will be converted to letter grades according to the following scale and cutoffs:

Final Average	[94, 100]	[90,94)	[84,90)	[80,84)	[77,80)	[74,77)	[70,73)	[67,70)	[64,67)	[60,64)	[57,60)	<57
Letter Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	E

Please be aware that a C- is not an acceptable grade for graduate students. A grade of C counts toward a graduate degree only if an equal number of credits in courses numbered 5000 or higher have been earned with an A.

Letter Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	E	WF	I	NG	S-U
Grade Points	4.0	3.67	3.33	3.0	2.67	2.33	2.0	1.67	1.33	1.0	0.67	0.0	0.0	0.0	0.0	0.0

For greater detail on the meaning of letter grades and university policies related to them, see the Registrar's Grade Policy regulations at: http://catalog.ufl.edu/ugrad/current/regulations/info/grades.aspx

Grade Response Times: grades will be posted as soon as possible after everyone has turned in their assignment. The exact turnaround time depends on the assignment's size and complexity.

Make-Up and Late-Submission Policy:

All work must be submitted via E-Learning by the exact due date and time (ET). You can review your graded work and our feedback by clicking "Grades" in the navigation panel.

- Late assignment submissions incur a 20% penalty per day (starting at 12:01am ET the day after the deadline).
- For the Final Interpretation Project, late work loses 10% per hour (beginning at 12:01am ET after it is due).
- Make-ups are ONLY permitted for illness or other unanticipated circumstances that cause a missed deadline., in line with college policy. A medical excuse and provider documentation are required.
- Work missed for any other reason receives a grade of zero.
- Technical-issue related make-ups request MUST be supported by the ticket number generated by LSS and an e-mail to the instructor within 24 hours of the problem.

Attendance Policy: This is an online course, "attendance" refers to routinely accessing the course materials, taking notes, contributing to the discussion boards, attending class meetings and office hours in Zoom. While assignments and quizzes have fixed deadlines to keep you on track, you are free to advance through the content if you need flexibility for future weeks. I encourage you, especially if you have questions or need help, to join my Zoom office hours/class meetings, these sessions are optional but particularly useful for clarifying concepts or demonstrating SAS code that are difficult to convey by message or email alone.

STUDENT EXPECTATIONS, ROLES, AND OPPORTUNITIES FOR INPUT

Communication Guidelines: Questions about course material can be posted on course discussion boards in E-Learning. Questions about specific quiz questions or issues of a personal nature will be sent by email through E-Learning. Please review the Netiquette Guidelines:

http://teach.ufl.edu/wp-content/uploads/2012/08/NetiquetteGuideforOnlineCourses.pdf.

Online Faculty Course Evaluation Process: Students are expected to provide professional and respectful feedback on the quality of instruction in this course by completing course evaluations online via GatorEvals. Guidance on how to give feedback in a professional and respectful manner is available at https://gatorevals.aa.ufl.edu/students/ . Students will be notified when the evaluation period opens, and can complete evaluations through the email they receive from GatorEvals, in their Canvas course menu under GatorEvals, or via https://ufl.bluera.com/ufl/. Summaries of course evaluation results are available to students at https://gatorevals.aa.ufl.edu/public-results/.

Academic Integrity: Students are expected to act in accordance with the University of Florida policy on academic integrity. As a student at the University of Florida, you have committed yourself to uphold the Honor Code, which includes the following pledge:

"**We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honesty and integrity**. "

You are expected to exhibit behavior consistent with this commitment to the UF academic community, and on all work submitted for credit at the University of Florida, the following pledge is either required or implied:

"On my honor, I have neither given nor received unauthorized aid in doing this assignment."

It is your individual responsibility to know and comply with all university policies and procedures regarding academic integrity and the Student Honor Code. Violations of the Honor Code at the University of Florida will not be tolerated. Violations will be reported to the Dean of Students Office for consideration of disciplinary action. For additional information regarding Academic Integrity, please see Student Conduct and Honor Code or the Graduate Student Website for additional details:

- https://www.dso.ufl.edu/sccr/process/student-conduct-honor-code/

Please remember cheating, lying, misrepresentation, or plagiarism in any form is unacceptable and inexcusable behavior.

ADVICE FROM DR. Wang

Do your best with the provided materials and to reach out whenever you need more direction or explanation.

- **Time commitment.** It is expected that you will spend approximately 9-12 hours per week on this course. This is roughly equivalent to 3 hours in class combined with 6-9 hours outside of class. Scheduling your time wisely and working efficiently will minimize the need for extra work in this course. Generally, we advise students to break this time up into small blocks split over as many days of the week as possible given your schedule. Working on too much material in one sitting is more likely to cause frustration and does not allow for time for understanding to develop or for questions to be answered.
- **Learn to use the materials to your greatest advantage.** If you ever feel lost or have issues, let us know immediately and we will help as soon as possible. Try to make sure as much of your time as possible is productive.
- **Learning is a process.** If you ever feel lost, please ask, but also understand that the course is building to a complete picture, and sometimes it's hard to see how each topic is related until later in the semester when we tie everything together. Often, activities and worksheets are leading you to think about things that will be important later in the course while working on skills related to the current topic.
- **Stay on track.** Try to make sure as much of your time as possible in the course is productive. Be sure to stay on track with the material and ask when you don't understand. **Getting behind can be difficult to fix in any course, but especially in an asynchronous one which is largely self-guided.** Let me know as soon as possible if you feel you are falling behind.

SUPPORT SERVICES

Accommodations for Students with Disabilities: If you require classroom accommodation because of a disability, you must register with the Dean of Students Office http://www.dso.ufl.edu within the first week of class. The Dean of Students Office will provide documentation to you, which you then give to the instructor when requesting accommodation. The College is committed to providing reasonable accommodations to assist students in their coursework.

Counseling and Student Health: Students sometimes experience stress from academic expectations and/or personal and interpersonal issues that may interfere with their academic performance. If you find yourself facing issues that have the potential to or are already negatively affecting your coursework, you are encouraged to talk with an instructor and/or seek help through University resources available to you.

- The Counseling and Wellness Center 352-392-1575 offers a variety of support services such as psychological assessment and intervention and assistance for math and test anxiety. Visit their web site for more information: http://www.counseling.ufl.edu. On line and in person assistance is available.
- You Matter We Care website: http://www.umatter.ufl.edu/. If you are feeling overwhelmed or stressed, you can reach out for help through the You Matter We Care website, which is staffed by Dean of Students and Counseling Center personnel.
- The Student Health Care Center at Shands is a satellite clinic of the main Student Health Care Center located on Fletcher Drive on campus. Student Health at Shands offers a variety of clinical services. The clinic is located on the second floor of the Dental Tower in the Health Science Center. For more information, contact the clinic at 392-0627 or check out the web site at: https://shcc.ufl.edu/
- Crisis intervention is always available 24/7 from:
  Alachua County Crisis Center: (352) 264-6789
  http://www.alachuacounty.us/DEPTS/CSS/CRISISCENTER/Pages/CrisisCenter.aspx

BUT – Do not wait until you reach a crisis to come in and talk with us. We have helped many students through stressful situations impacting their academic performance. You are not alone so please do not be afraid to ask for assistance.

Pacing Guide for Course Materials

For more details, see E-Learning. Due dates for assignments will be provided in the Course website on Elearning.

Week	Monday	Topic
1	1/12	**Unit 1:** Introduction & Review
2	1/19	**Unit 1:** Introduction & Review
3	1/26	**Unit 2:** Simple Linear Regression
4	2/2	**Unit 2:** Simple Linear Regression
5	2/9	**Unit 3:** Multiple Linear Regression
6	2/16	**Unit 3:** Multiple Linear Regression
7	2/23	**Unit 3:** Multiple Linear Regression
8	3/2	**Unit 3:** Multiple Linear Regression
9	3/9	**Unit 4:** Simple Logistic Regression
		Spring Break!
10	3/23	**Unit 5:** Multiple Logistic Regression
11	3/30	**Unit 5:** Multiple Logistic Regression
12	4/6	**Unit 5:** Model Selection
13	4/13	**Unit 6:** Model Selection
14	4/20	**Unit 7:** Poisson Regression & Generalized Linear Models (GLMs)
15	4/27	Finals Week

Disclaimer: The instructor reserves the right to modify this syllabus at her discretion as needed.